5.1.3 Neuronal communication

The stimulation of sensory receptors leads to the generation of an action potential in a neurone.

Transmission between neurones takes place at synapses.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the roles of mammalian sensory receptors in converting different types of stimuli into nerve impulses	To include an outline of the roles of sensory receptors (e.g. Pacinian corpuscle) in responding to specific types of stimuli and their roles as transducers.
(b)	the structure and functions of sensory, relay and motor neurones	To include differences between the structure and function of myelinated and non-myelinated neurones.
(c)	the generation and transmission of nerve impulses in mammals	To include how the resting potential is established and maintained and how an action potential is generated (including reference to positive feedback) and transmitted in a myelinated neurone **AND** the significance of the frequency of impulse transmission. *M1.3, M3.1*
(d)	the structure and roles of synapses in neurotransmission.	To include: • the structure of a cholinergic synapse • the action of neurotransmitters at the synapse • the importance of synapses in summation • inhibitory and excitatory synapses.